



PHOTO BY UMESH GOSWAMI

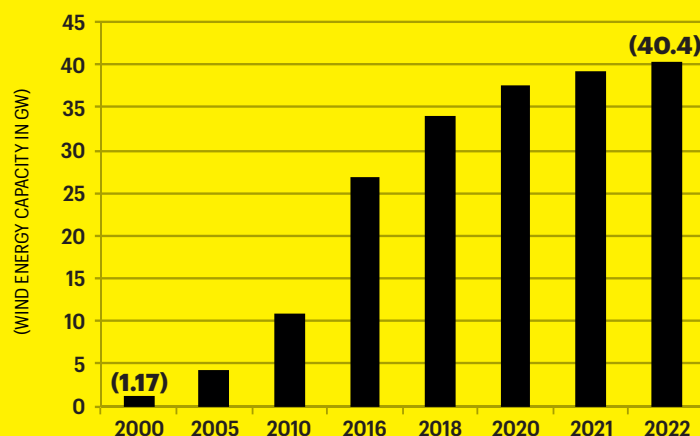
▶ **To reach net-zero emissions by 2070, \$160 billion per year is needed on average, across India's energy value chain between now and 2030**

▶ **India may need a total of \$17.77 trillion, including \$12.4 trillion in investments, to meet its 2070 net-zero goal**

▶ **The country may thus need to draw a road map for accessing low-cost long-term capital**

WINDFALL

Capacity addition in India's wind energy sector has slowed a bit in recent times



SOURCE: MINISTRY OF POWER AND MINISTRY OF NEW & RENEWABLE ENERGY

ied on Chinese imports. In August 2022, the country's power systems operator, the Power System Operation Corporation (POSOCO) barred 27 power distributors from buying electricity from power exchanges to meet their short-term requirements over non-payment of dues to generators. Simultaneously, the Electricity (Amendment) Bill, 2022, that seeks to bring about comprehensive reform is currently under review by a parliamentary committee. More than 93.4 million LPG connections had been released till July 2022 for below poverty line (BPL) households under the Pradhan Mantri Ujjwala Yojana (PMUY) for cooking.

The country has also been trying to bring down the cost of storage of renewable energy. "You can't have energy transition unless you crack the storage problem. Storage has to become viable. All these developed countries that have been talking about the need for climate action

and energy transition have done nothing. The largest quantum of storage set up by them is 450 MW!" says Power Minister Singh. He says the disruption caused to global supply chains in the aftermath of the Covid-19 pandemic was a lesson for the world and every effort must be made to expand them by moving away from lithium-ion to other battery chemistries. In April 2022, the Solar Energy Corporation of India, a public sector enterprise under the Ministry of New & Renewable En-

ergy, came out with a tender for an aggregate storage capacity of 1,000 MWh, the largest such bid globally.

Honeywell's Gaikwad is confident of India's potential to deliver. "You will now see the announcement of giga factories... Giga factories for making solar panels, fuel cells, electrolyzers and batteries; all of these components are essential for energy transition. These will be probably the world's biggest such factories [in India] and will come about in a span of the next two to three years," he declares.

HYDROGEN HUB

On January 4, India announced the National Green Hydrogen Mission. Power Minister Singh forecasts India's emergence as the commodity's leading supplier. "The [production] capacity for green hydrogen will go up from 5 million metric tonnes (MMT) to at least 10 MMT [by 2030] because we had started putting in

10 MMT

India's green hydrogen production capacity target in 2030, compared to the current level of 5 MMT